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Customer Churn Prediction for Telecom Companies

Capstone project 1: Building a machine learning model to predict telecom customer churn and help companies retain subscribers.

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Problem Statement

Customer churn — the voluntary discontinuation of a service — represents one of the most costly challenges facing telecommunications providers. Retaining an existing customer costs significantly less than acquiring a new one, yet churn rates in telecom can reach **20–30% annually**.

Why Churn Hurts

- Direct revenue loss from departing subscribers
- Increased customer acquisition cost (CAC) to compensate
- Erosion of brand loyalty and market position

What Businesses Must Do

- Proactively identify customers at high risk of churning
- Intervene with targeted retention strategies before churn occurs
- Prioritize resource allocation toward highest-risk segments



Project Objectives

This project delivers a rigorous, end-to-end machine learning pipeline designed to predict telecom customer churn with high precision and interpretability.

1	Predictive Modeling Build and train a robust ML classifier to accurately predict binary churn outcomes.
2	Imbalanced Data Handling Address class imbalance using SMOTE and other resampling techniques for unbiased learning.
3	Model Comparison Benchmark multiple classification algorithms to identify the highest-performing approach.
4	Robust Evaluation Assess models using ROC-AUC, Recall, F1-Score — metrics suited for imbalanced classification.
5	Explainability Provide SHAP-based feature importance insights for actionable business interpretation.
6	Deployment Deliver an interactive Streamlit application for real-time churn scoring and business use.

Dataset Overview

The project utilizes the publicly available **IBM Telco Customer Churn** dataset — a widely adopted benchmark in churn analytics research, containing rich behavioral, demographic, and financial attributes.

7,043

Total Records

Unique customer entries representing a diverse subscriber base

21

Feature Columns

Spanning demographic, service, account, and financial dimensions

~26%

Churn Rate

Positive class (Churn_Value = 1) representing a significant class imbalance challenge



Target Variable: Churn_Value — Binary label where 0 = customer retained and 1 = customer churned. Provided in CSV/Excel format, ready for direct ingestion into the ML pipeline.

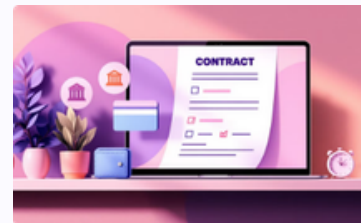
Feature Categories

Features are organized into four semantically meaningful groups, enabling structured preprocessing, targeted feature engineering, and domain-aligned model interpretability.



Demographic Features

- Gender
- Senior Citizen status
- Partner & Dependents



Account Information

- Tenure (months)
- Contract Type
- Payment Method



Financial Features

- Monthly Charges
- Total Charges
- Customer Lifetime Value (CLTV)



Service Features

- Internet Service type
- Online Security
- Tech Support subscription

Project Workflow

The pipeline follows a structured, reproducible sequence from raw data ingestion through model deployment — ensuring methodological rigor at each stage.



Each stage is modular and independently testable, enabling iterative refinement without disrupting downstream components. SMOTE is applied exclusively on the training set to prevent data leakage.

Data Cleaning & Preprocessing

A comprehensive cleaning pipeline was applied to ensure data integrity and consistency before any modeling steps were performed. All transformations were documented and version-controlled.

Structural Fixes

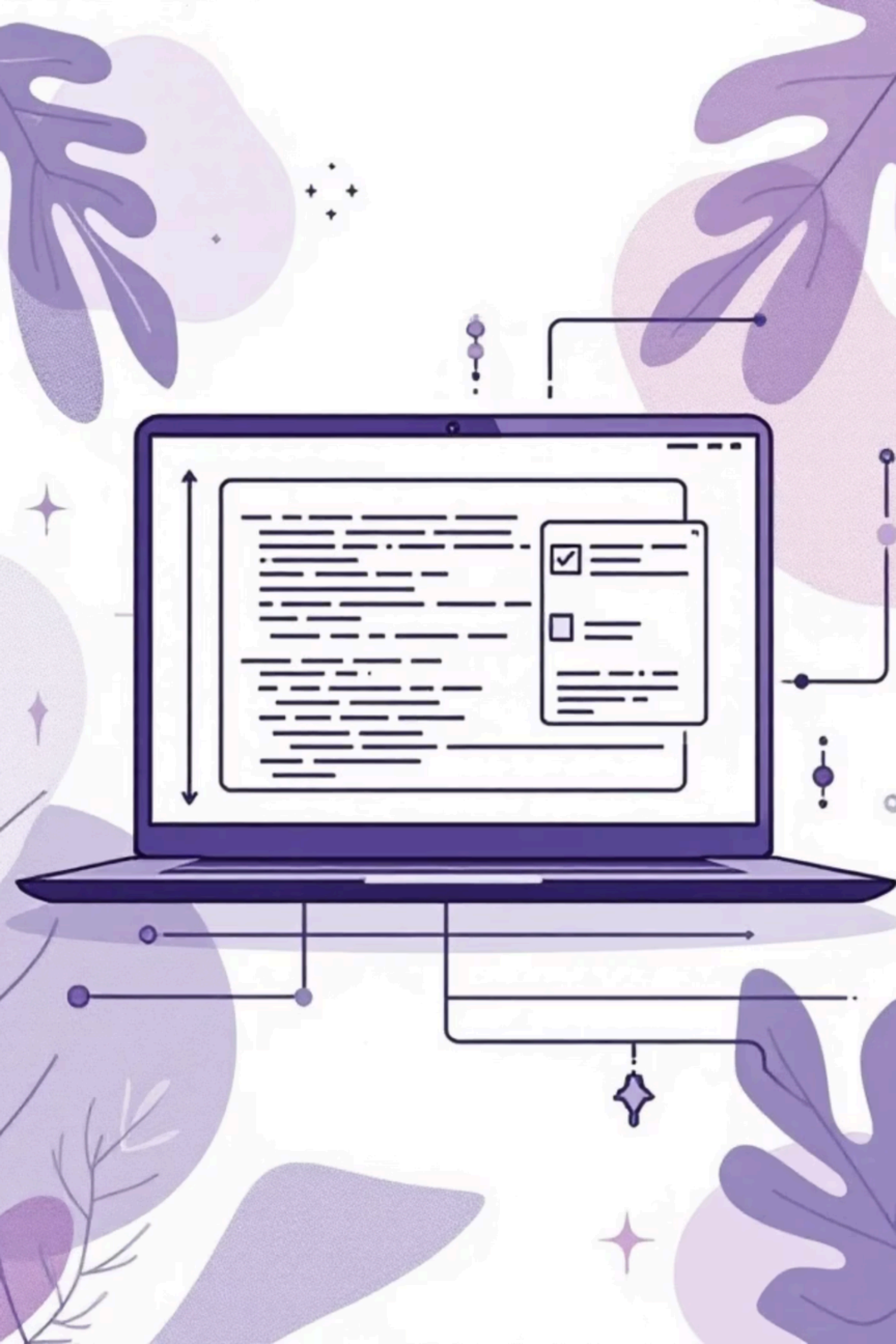
- Dropped irrelevant and leakage columns: CustomerID, Lat Long, Zip Code, City, State, Country, Churn Reason, Count.
- Standardized column names by replacing spaces with underscores (e.g., Total Charges → Total_Charges).
- Converted Total_Charges from string to numeric using coercion, ensuring type consistency.

Missing Value Strategy

- For Total_Charges, missing values were imputed using the median of the column, preserving robustness against outliers.

Outcome

Dataset cleaned of irrelevant identifiers and leakage features. Column names standardized for consistency. Total_Charges is now numeric, with no missing values. Ready for downstream feature engineering and model training.



Feature Engineering

Target Variable

The model predicts **Churn_Value** — a binary indicator of whether a customer has churned. This serves as the ground truth label for all supervised learning tasks.

Data Leakage Prevention

Two features were removed prior to training to prevent data leakage — columns that would not be available at prediction time in a real-world deployment:

- **Churn_Label** — direct text encoding of the target
- **Churn_Score** — a pre-computed churn probability

One-Hot Encoding

Categorical variables were transformed using `pd.get_dummies()`, converting nominal features into binary indicator columns compatible with all classifiers.

Cardinality Reduction

High-cardinality text columns (e.g., customer IDs, free-text fields) were dropped to reduce noise and dimensionality in the feature space.

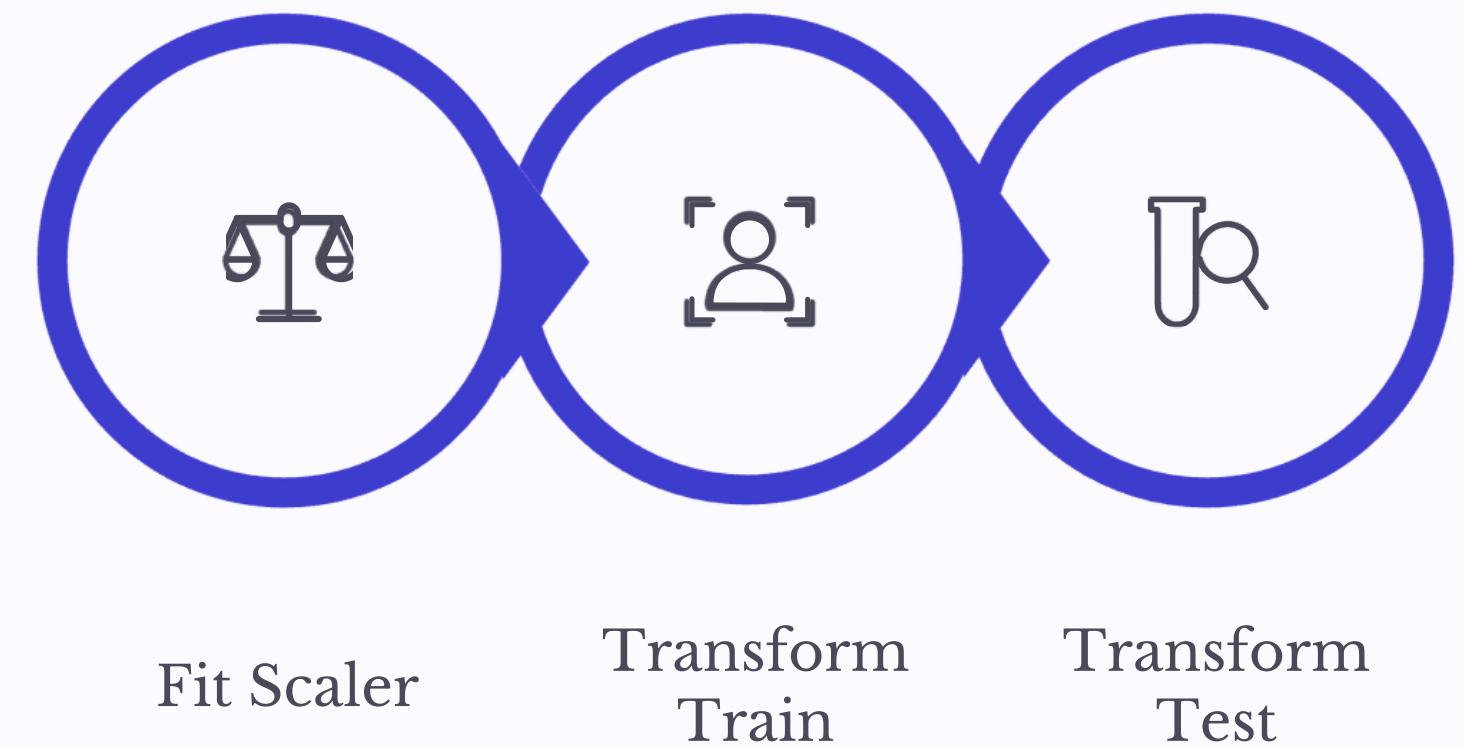
Train-Test Split & Feature Scaling

Stratified 80/20 Split

The dataset was divided into **80% training** and **20% testing** subsets. Stratified sampling was applied to ensure both splits maintain the same churn-to-non-churn class ratio as the full dataset — critical for reliable evaluation on imbalanced data.

StandardScaler Workflow

Feature scaling was applied using **StandardScaler** to normalize numerical features to zero mean and unit variance. The scaler was *fit exclusively on training data* and subsequently used to transform the test set, preventing any information leakage from the test distribution into model training.



Handling Imbalanced Data with SMOTE

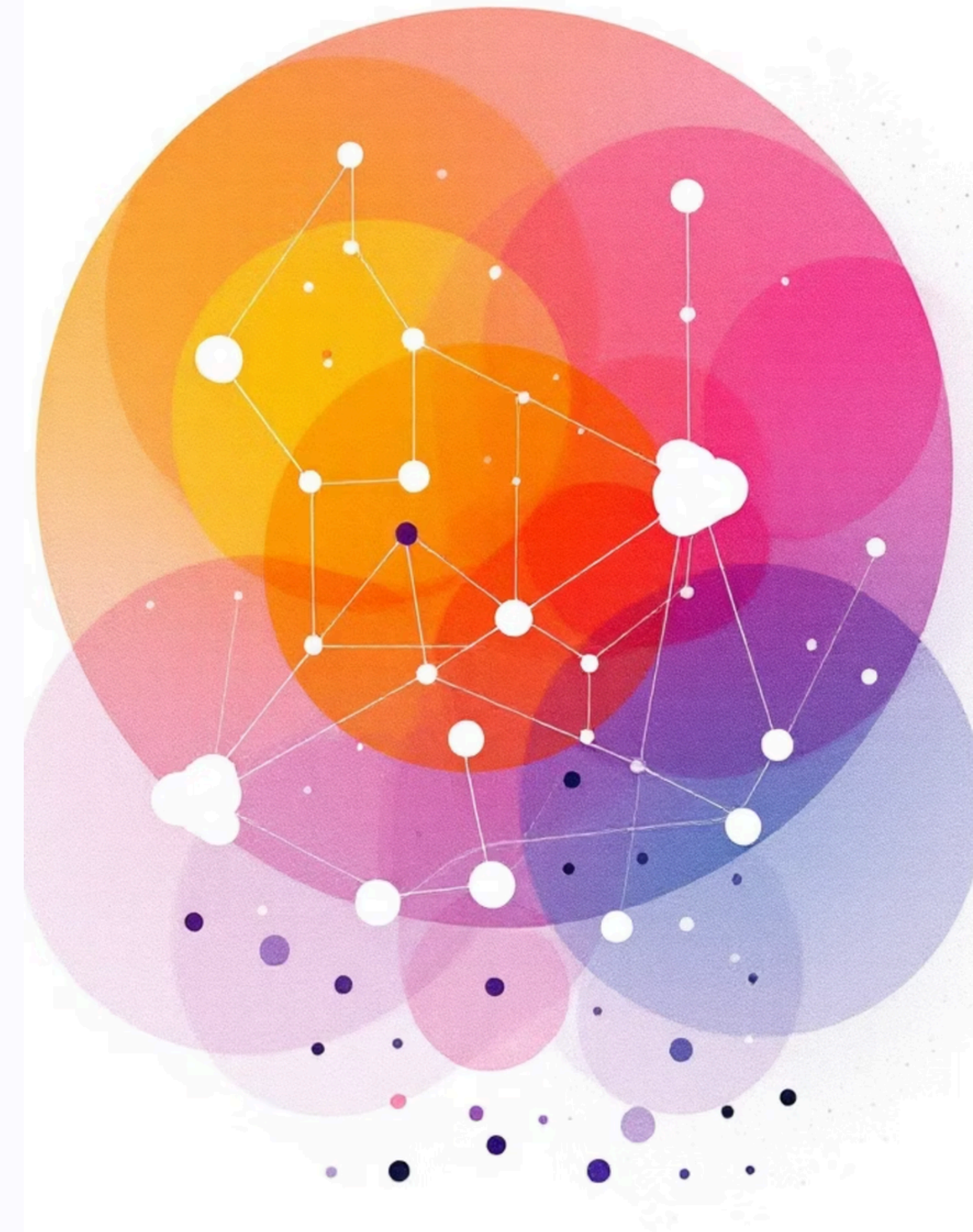
The Problem

Churn events are inherently rare relative to the overall customer base. A naïve model trained on imbalanced data will be biased toward the majority class, achieving high accuracy by simply predicting "no churn" while failing to detect actual churners.

SMOTE: Synthetic Minority Oversampling Technique

SMOTE addresses class imbalance by **generating synthetic samples** for the minority (churn) class. Rather than duplicating existing records, it interpolates between neighboring minority-class instances in feature space, creating plausible new examples. This enriches the training signal and improves the model's ability to learn churn boundaries.

❏ **Important:** SMOTE was applied *only to the training dataset* — the test set retains its original distribution to ensure unbiased evaluation.



Models Used

Four classifiers were selected to cover a spectrum of complexity, interpretability, and ensemble strategy — enabling robust cross-model comparison.



Random Forest

An ensemble of decorrelated decision trees trained via bagging. Naturally handles non-linear relationships and feature interactions. Robust to overfitting and provides built-in feature importance scores.



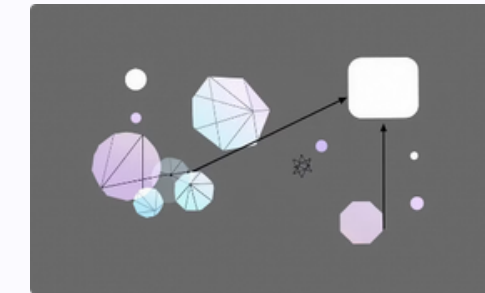
XGBoost

A high-performance gradient boosting framework that builds trees sequentially, correcting prior errors. Known for strong accuracy on structured/tabular data and efficient handling of sparse features.



Logistic Regression

A simple, interpretable linear baseline model. Provides probability calibration and clear coefficient-based insights. Essential for benchmarking more complex models against a transparent baseline.



AdaBoost

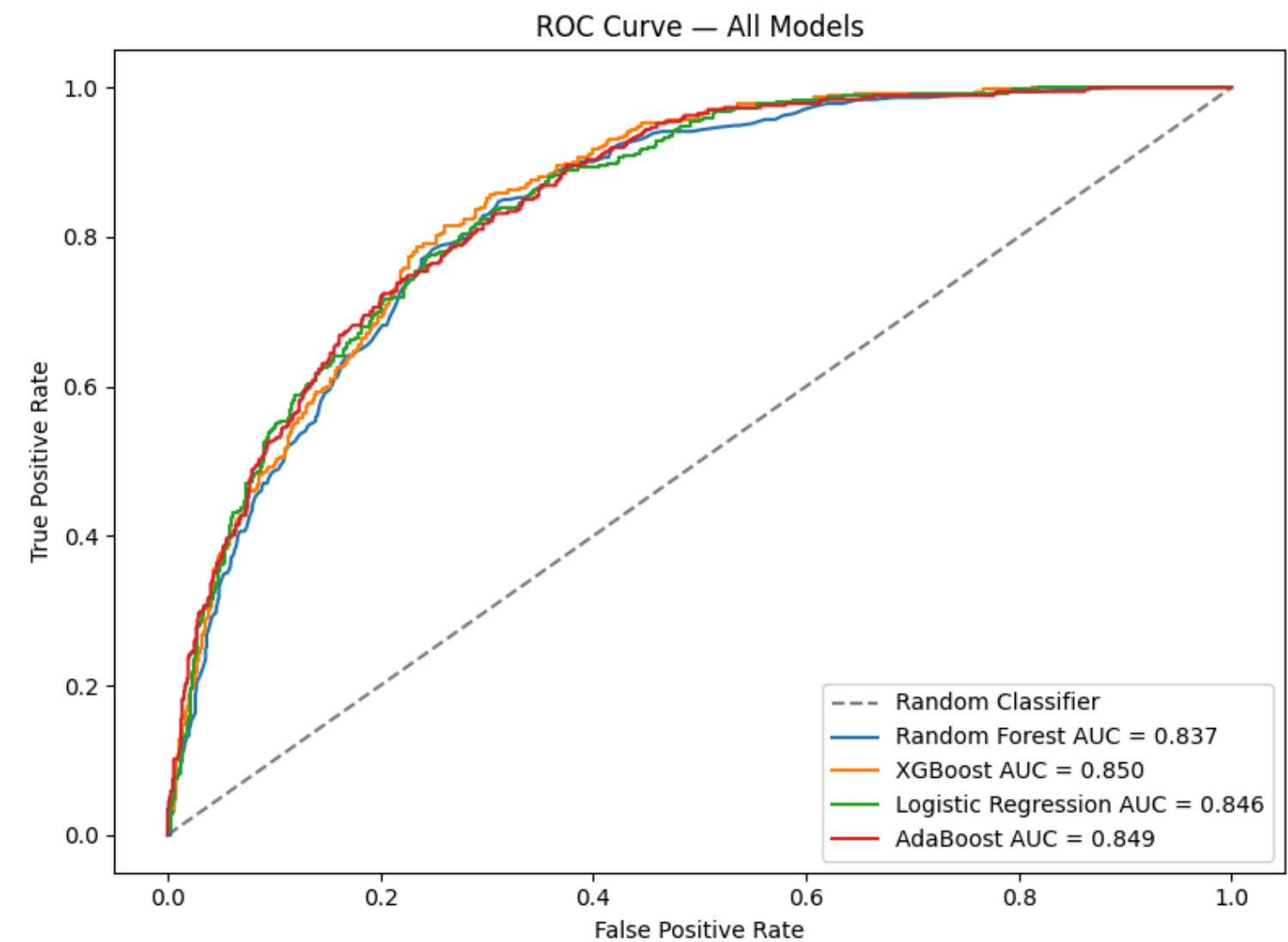
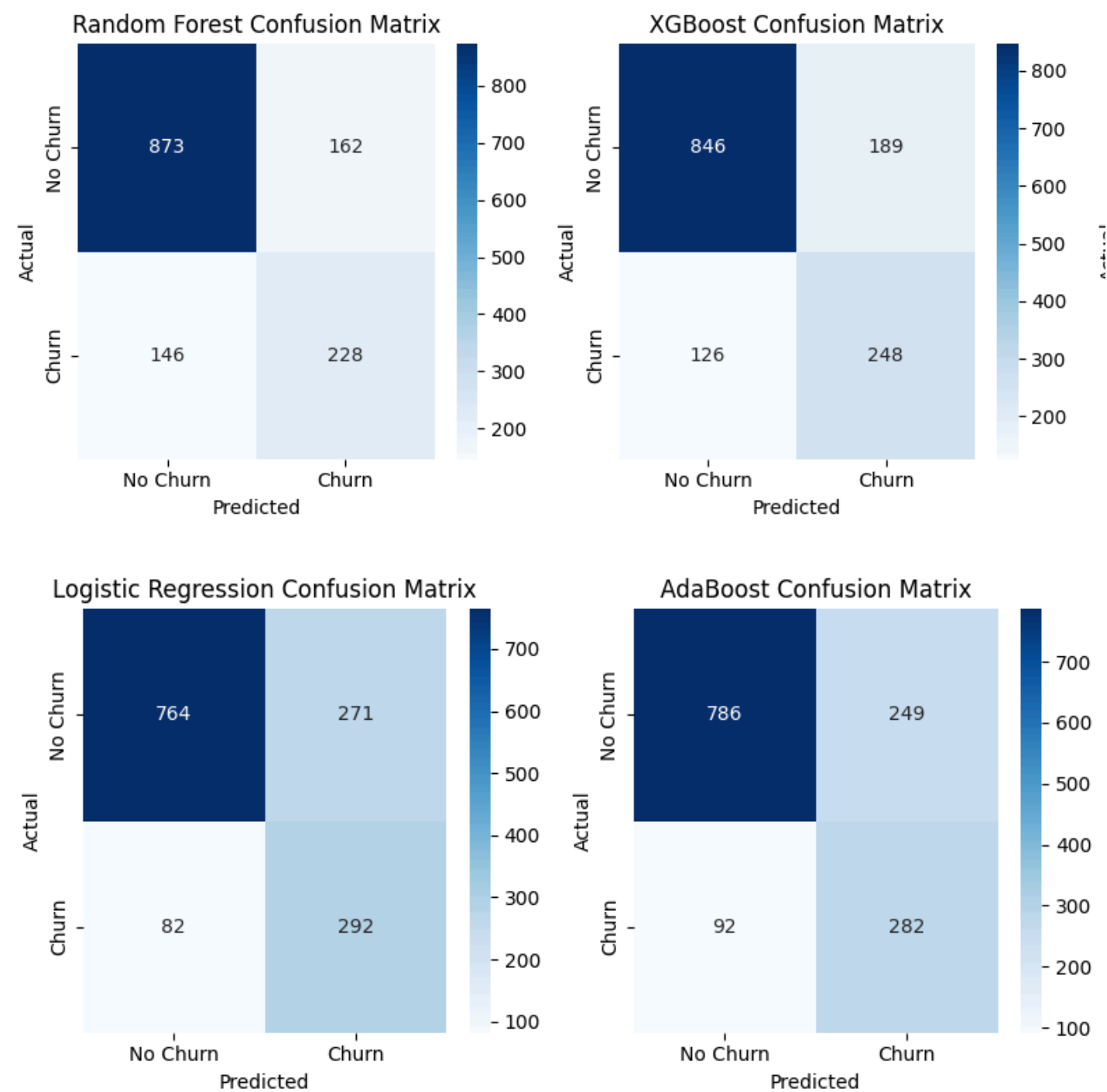
An adaptive boosting technique that iteratively focuses on misclassified instances by upweighting difficult examples. Combines multiple weak learners into a strong composite classifier.

Model Evaluation Metrics

A comprehensive set of metrics was selected to capture different dimensions of classifier performance — especially important given the class imbalance in churn detection.

1	Accuracy Proportion of correct predictions overall. Useful as a high-level indicator but can be misleading under class imbalance.
2	ROC-AUC Primary metric. Measures the model's ability to rank churners above non-churners across all classification thresholds. Threshold-invariant and robust to imbalance.
3	Precision Of all customers flagged as churners, what fraction truly churned? Minimizes false alarms and wasted retention spend.
4	Recall Of all actual churners, what fraction did the model catch? High recall is critical to ensuring at-risk customers are not missed.
5	F1 Score Harmonic mean of Precision and Recall. Balances both concerns into a single metric, ideal when both false positives and false negatives carry cost.
6	Confusion Matrix Breaks predictions into TP, FP, TN, and FN quadrants — providing granular insight into error types beyond aggregate summary statistics.

Model Evaluvation



Model	Accuracy	Precision	Recall	F1-Score
Random Forest	~78%	~58%	~61%	~60%
XGBoost	~78%	~57%	~66%	~61%
Logistic Regression	~75%	~52%	~78%	~62%
AdaBoost Classifier	~76%	~53%	~75%	~62%

*In churn prediction, recall is usually prioritised because missing churners is more costly than targeting a few extra non-churners. That means Logistic Regression or AdaBoost will be the most useful.

Model Performance Results

All four models were evaluated on the held-out test set. XGBoost achieved the best overall ranking ability, while Logistic Regression demonstrated strong churn recall.

Random Forest Accuracy ~78% Strong and consistent baseline performance. Handles feature interactions well, with competitive scores across all metrics. Reliable choice for production deployment.	XGBoost Best ROC-AUC Score Top-ranked model for discriminative ability. Excellent balance of precision and recall. Best suited as the primary production model for churn scoring.
Logistic Regression Recall ~78% Highest recall among all models — detects the largest share of actual churners. Preferred when the cost of missing a churner exceeds the cost of a false positive.	AdaBoost Balanced Performance Consistent scores across accuracy, precision, recall, and F1. A reliable ensemble option, especially when no single metric dominates the business objective.

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Explainability & Key Churn Drivers

SHAP: SHapley Additive Explanations

SHAP was used to quantify each feature's contribution to individual predictions. Rooted in cooperative game theory, SHAP values provide a theoretically principled, consistent attribution of model output — enabling both global feature ranking and local per-customer explanation.

Top Identified Churn Drivers

- **Month-to-month contracts** — lowest commitment, highest churn risk
- **Low tenure** — new customers churn disproportionately often
- **High monthly charges** — price sensitivity drives disengagement
- **No tech support or online security** — lack of stickiness services
- **Electronic payment method** — associated with transactional, low-loyalty behavior

Deployment & Conclusion

Streamlit Application

The trained pipeline was deployed as an interactive **Streamlit web application**, enabling non-technical stakeholders to operationalize churn prediction without writing code. The app supports three core workflows:

01

Upload Dataset

Ingest new customer data in CSV format with automated validation and preprocessing.

02

Train Models

Trigger the full ML pipeline — feature engineering, SMOTE, scaling, and model fitting — on demand.

03

Predict Churn

Generate individual churn probabilities and batch scores with SHAP-based explanation output.

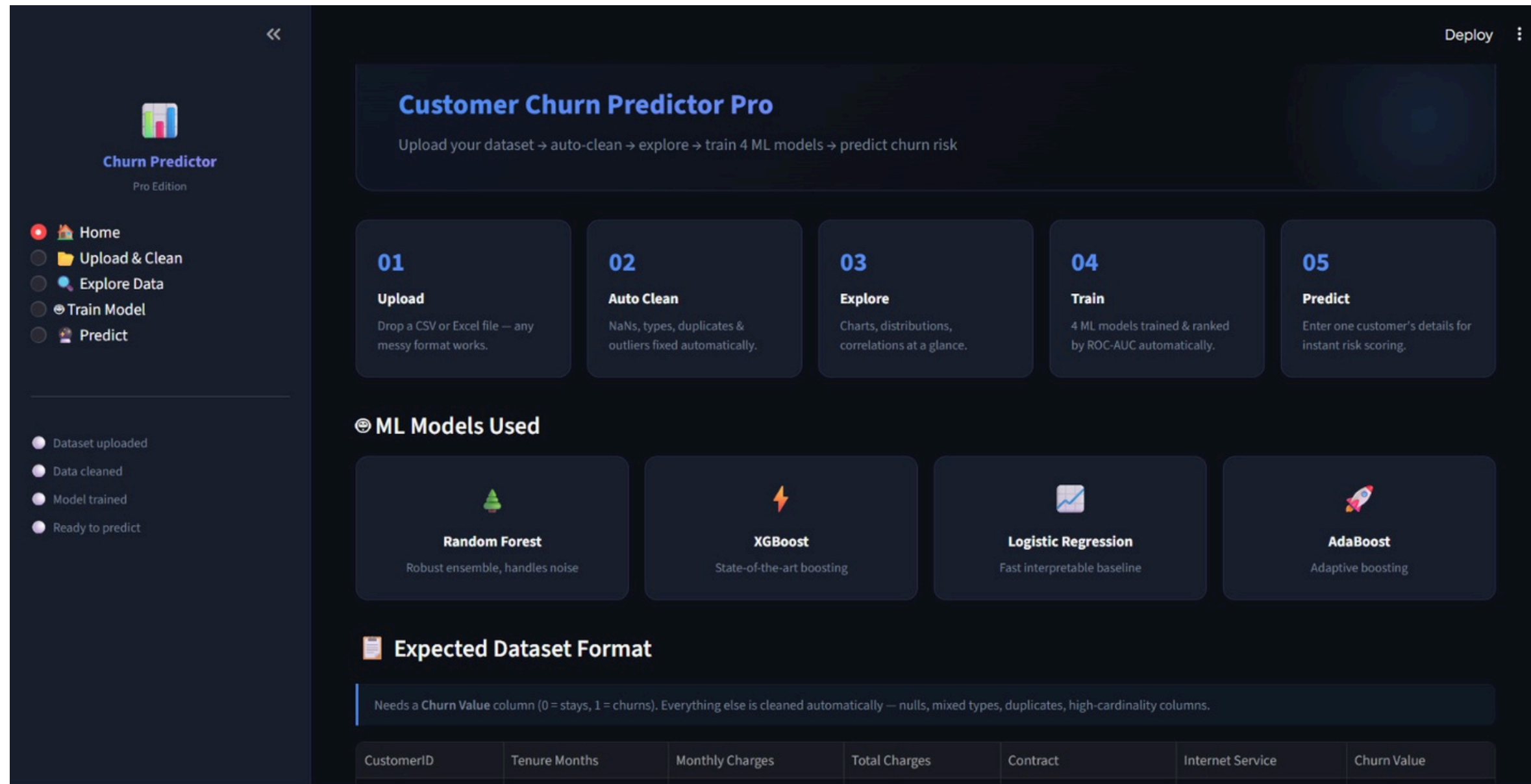
Key Outcomes & Conclusion

This project delivers a complete, production-ready **end-to-end ML pipeline** for telecom customer churn prediction. Key outcomes include:

- Accurate churn classification across multiple model architectures
- Business-interpretable insights via SHAP feature attribution
- Actionable retention signals tied to specific customer behaviors
- Accessible deployment enabling non-technical adoption

The system empowers organizations to shift from *reactive* to **proactive** customer retention — identifying at-risk customers before they disengage.

Light version of the app



Live Deployment

Streamlit Cloud: <https://customerchurnprediction-23z6zwjak8kgu2sktpvqrm.streamlit.app/>



Thank You...